Professional Certificate Programme in Agentic AI and Applications

Real-World Applications & Case Studies

Revisit the Learning So Far

Let's take a quick look at the key ideas and concepts you have explored in your learning journey.

- Role of ethics, safety and governance in guiding agentic AI across diverse contexts
- Role of responsible AI as a continuous process of monitoring, evaluation and improvement
- Identification and management of agent-specific risks such as bias, hallucination and misuse
- Importance of privacy, consent and regulatory compliance in autonomous systems
- Need for guardrails, transparency and human oversight to balance autonomy and safety

Topics Covered

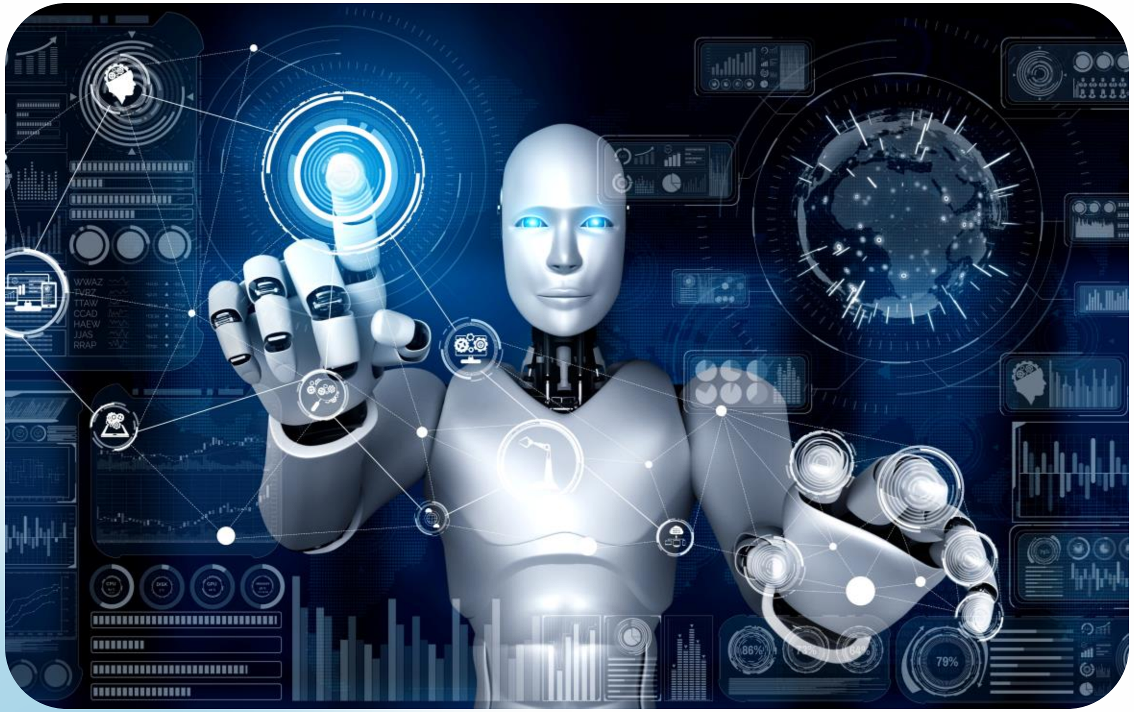
- AI Agents in the Real World
- Real-World AI Agents
- ROI of Agent Adoption
- Success Stories & Failures
- Business Workflow Automation Agent
- Domain-Specific Brainstorm
- Real-World Case Studies

AI Agents in the Real World
Impact, ROI & Use Cases

Transforming industries through autonomous intelligence

AI Agents in the Real World

Impact, ROI & Use Cases



- **From automation**
  Traditional scripted workflows
- **To autonomy**
  Decision-making intelligence
- **Real impact**
  Measurable business outcomes

Instructor-led exploration with real-world case studies and interactive discussion

Session Objectives

By the End of this Session, You Will:

- **Master real-world applications:** Understand AI agent implementations across industries
- **Evaluate ROI frameworks:** Learn measurement methods with case examples
- **Analyse success patterns:** Study winning strategies and avoid failure points
- **Generate domain ideas:** Brainstorm organisation-specific use cases
- **Interactive engagement:** Participate in collaborative Q&A and peer discussions

Types of AI Agents

Understanding the Spectrum of Agent Architectures and Capabilities

Reactive agents
Respond instantly to stimuli without memory or planning

Proactive agents
Anticipate future states and plan multi-step actions

Conversational agents
Engage users through natural, contextual dialogue

Types of AI Agents

Understanding the Spectrum of Agent Architectures and Capabilities

Reasoning agents
Solve complex problems with multi-step inference

Multi-agent systems
Coordinate specialised roles to achieve complex goals

Why Agents Matter Now

24/7
Always-on operation
Continuous reasoning and contextual decision-making

10x
Productivity gains
Automation beyond simple rule-based workflows

∞
Scalable intelligence
Foundation for next-gen enterprise capabilities

Why Agents Matter Now

The Perfect Storm

- LLM-powered reasoning at scale
- Seamless API and tool integration
- Real-time contextual adaptation
- Enterprise-grade accuracy thresholds

Real-World AI Agents

From theory to practice: agents transforming industries today

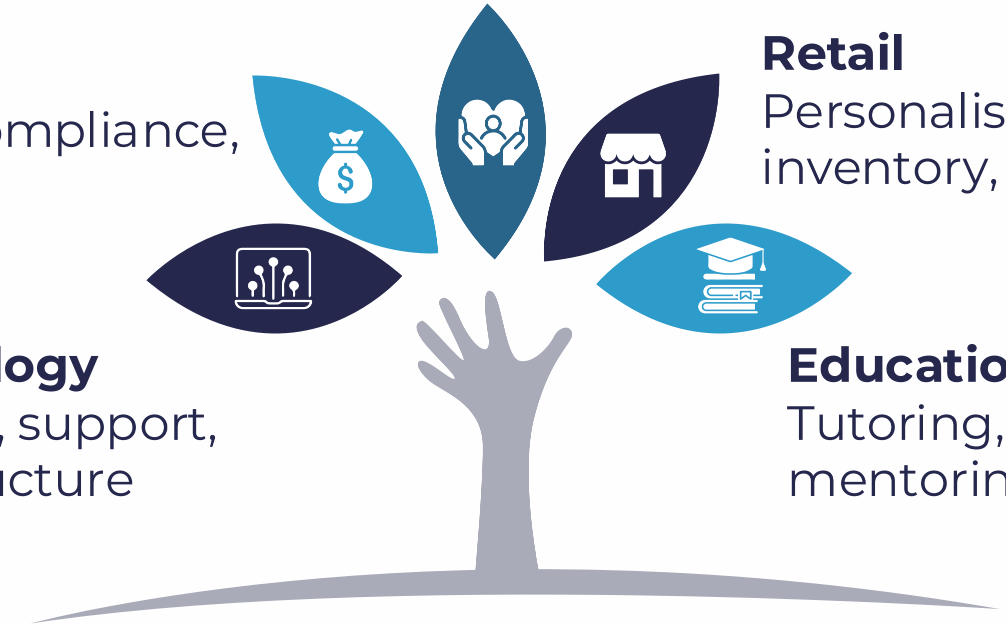
Industry Adoption Overview

AI Agents are Embedded Across Verticals, Driving Unified Outcomes

Healthcare
Triage, scheduling, follow-up

Finance
Fraud, compliance, advisory

Retail
Personalisation, inventory, CX



Technology
DevOps, support, infrastructure

Education
Tutoring, grading, mentoring

Common goals: Reduce operational costs, improve response speed, personalise service delivery at scale

Customer Service Agents

Intelligent Support at Scale

Platforms like **Intercom, Ada and ChatGPT CX** handle complex support workflows autonomously.

Automated resolution
Handle FAQs, refunds and order tracking without human escalation

Contextual learning
Analyse past tickets to personalise tone and recommendations

24/7 availability
Provide instant responses across time zones and languages

Healthcare Agents

Virtual triage
Deliver symptom assessment and care routing through platforms like **Babylon** and **Ada Health**

Patient engagement
Support patients with automated follow-ups, medication reminders and wellness check-ins

Administrative relief
Reduce appointment coordination overhead by up to 70% with AI-based scheduling

Improving patient outcomes while reducing clinical burnout

Manufacturing & IoT Agents

Intelligent Operations

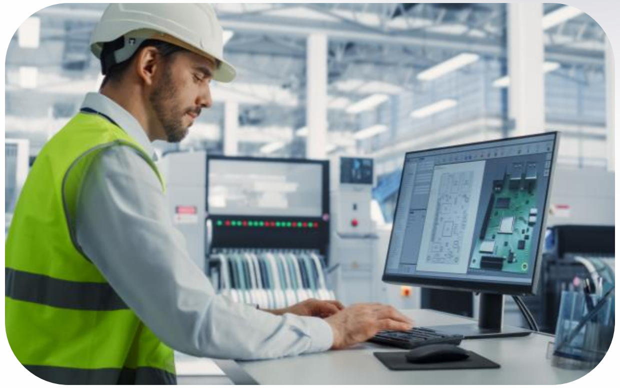
Real-time sensor analysis and autonomous decision-making transform production floors.

Predictive maintenance

Analyse machine data to predict failures before downtime

Production optimisation

Allocate resources dynamically using real-time conditions



Intelligent alerting

Deliver proactive notifications to prevent costly production stoppages

Finance & Banking Agents

Fraud prevention

Detect fraud in real time and automate anti–money laundering workflows

Wealth advisory

Provide portfolio management and chat-based credit scoring through AI

Compliance automation

Process loan documentation and automate regulatory reporting

Result: Faster decisions, reduced risk exposure, enhanced customer trust

Retail & E-commerce Agents

Personal shopping
Guide purchase decisions through conversational AI using assistants such as Shopify Sidekick

Smart recommendations
Deliver context-aware product suggestions via natural dialogue interfaces

Order management
Automate returns, status updates and fulfilment tracking

Personalisation at scale
Convert browsers into buyers through intelligent engagement

Supply Chain Agents

Inventory monitoring

Track dynamic stock levels across multiple locations

Logistics coordination

Optimise routing and carrier allocation through multi-agent systems

Demand prediction

Analyse real-time signals to drive proactive replenishment

End-to-end visibility and autonomous decision-making across the value chain

Education & Training Agents

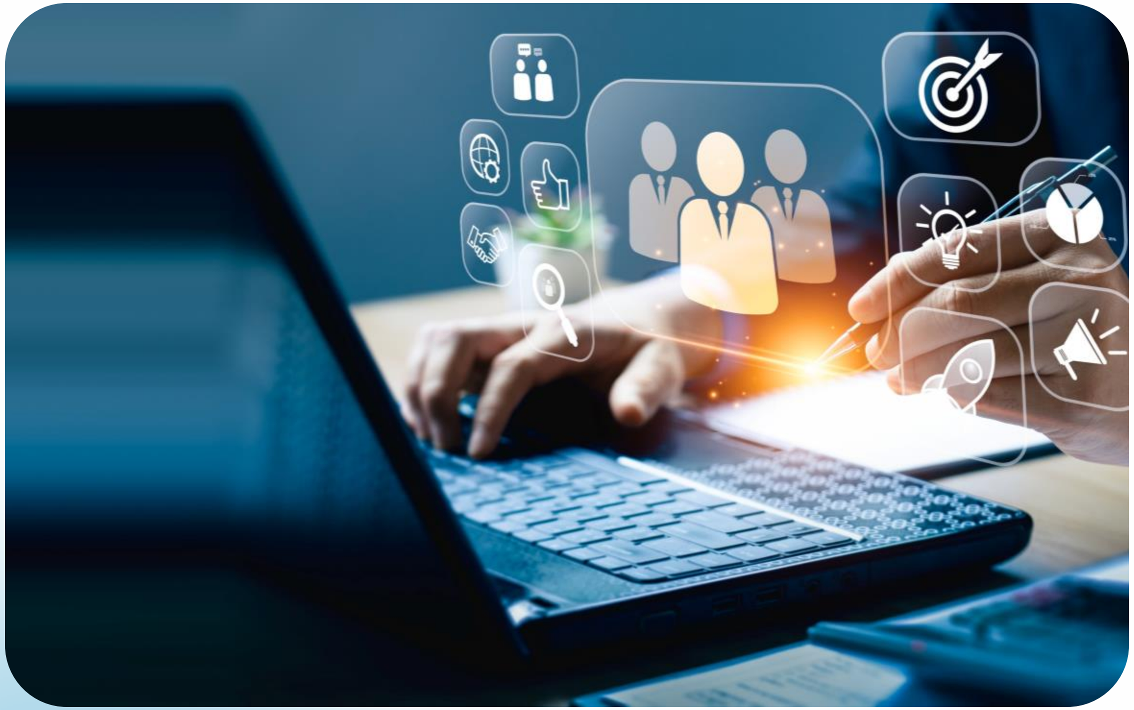
Adaptive Learning Systems

Personalised tutoring
Adapt to each learner's pace and style on platforms like **Duolingo Max** and **Khanmigo**

Real-time feedback
Provide instant correction and encouragement during practice

Assessment automation
Grade essays at scale while delivering constructive mentoring

HR & Recruitment Agents



Resume screening
Use sentiment-based candidate shortlisting and skills-matching

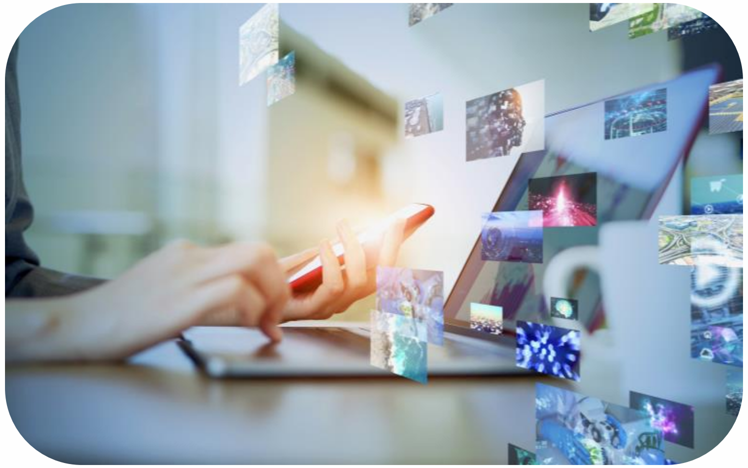
Employee support
Handle queries, provide policy guidance and power onboarding helpdesks

Continuous development
Deliver personalised learning nudges and career-pathing recommendations

Impact: Faster hiring cycles, improved candidate experience, reduced administrative burden

Creative & Marketing Agents

Use tools like **Jasper** and **Copy.ai** to generate creative concepts and messaging variants



Use multi-agent systems to manage social post scheduling and cross-channel coordination

Apply data-driven A/B testing and campaign iteration at machine speed

Creative amplification: Agents don't replace human creativity—they scale it, enabling marketers to test more ideas faster.

Section 3 –ROI of Agent Adoption

Business Value Dimensions

AI agents deliver measurable impact across four critical business pillars

Cost efficiency
Reduce manual touchpoints and operational overhead

Speed
Process routine tasks instantly without delays

Revenue growth
Unlock upselling and cross-selling at scale

Customer experience
Deliver personalisation that scales effortlessly

ROI Formula for AI Agents

The formula: ROI = $\frac{(Benefits - Costs)}{Costs} \times 100$

Benefits
Generate cost savings + revenue gains from automation

Costs
Account for infrastructure, training and integration expenses

Timeline
Track performance outcomes over a 6–12-month deployment period

Quantitative Impact Metrics

Key Performance Indicators that Prove AI Agent Effectiveness

Task automation rate (%)
Percentage of workflows handled without human intervention

Average resolution time reduction
Minutes or hours saved per customer interaction

Agent utilisation ratio
Optimal deployment across concurrent workflows

Cost saved per transaction
Direct financial impact per automated process

ROI Benchmarks by Sector

Real-World Performance Data Across Major Industries

30-40%

Customer support

Cost reduction through automated resolution

20%

Manufacturing

Downtime reduction via predictive maintenance

15-25%

Finance

Fraud loss prevention through intelligent detection

3 x

Retail

Improvement in customer engagement rates

ROI Measurement Challenges

Attribution complexity
Isolate the agent impact from broader AI stack contributions

Soft benefits
Quantify intangible outcomes such as customer satisfaction and brand loyalty

Data maturity
Establish accurate performance baselines through a robust data infrastructure

Hidden displacement
Identify shadow labour shifts that traditional metrics often fail to capture

Success Stories & Failures

Learning from both triumphs and setbacks in AI agent deployment

Case Study 1 – Amazon's Supply Chain Agents

Intelligent Logistics at Scale

Dynamic routing: Agents predict delivery bottlenecks before they occur

Ai-led procurement: Autonomous planning optimises inventory levels

Measurable impact : 25% faster fulfilment with significantly reduced excess stock

Case Study 2 – Duolingo Max

GPT-powered tutoring: Conversational practice adapts to learner needs in real-time

Personalised feedback : Every learner receives customised guidance and corrections

Breakthrough results : **40% increase** in learning engagement metrics

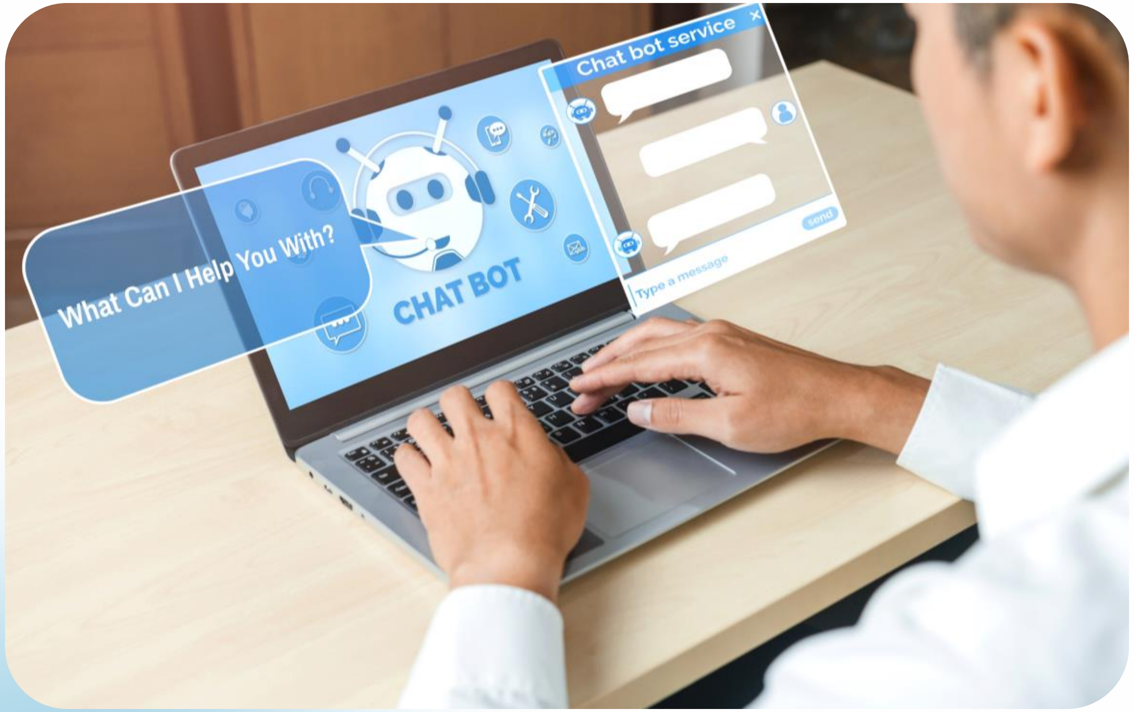
Case Study 3 – Microsoft Tay Bot (Failure)

What went wrong: Unfiltered social learning caused offensive behaviour within 24 hours of launch

Critical lesson: Guardrails and continuous supervision are non-negotiable

Ethics must always take precedence over speed of deployment

Case Study 4 – Banking Chatbot Misinterpretation



The incident: Misleading financial suggestions triggered compliance violations

Root cause: Insufficient domain-trained context and audit log capabilities

Critical takeaway : Rigorous testing and validation required before customer-facing deployment

Trust is earned through comprehensive validation, not speed to market

Lessons from Failures

Common Failure Patterns Reveal Essential Deployment Requirements

Misalignment

Disconnect between agent intent and actual actions taken

Overfitting

Excessive reliance on conversational data without context validation

Missing oversight

No human-in-the-loop safeguards for critical decisions

Content gaps

Lack of real-time filtering and monitoring mechanisms

Success Patterns

Clear KPIs defined pre-deployment
Set measurable success criteria before launch

Feedback-driven learning loops
Improve continuously based on real-world performance data

Gradual rollout with manual supervision
Reduce risk through phased deployment with human oversight

Integration with existing workflows
Enable seamless adoption within current operational infrastructure

Business Workflow Automation Agent

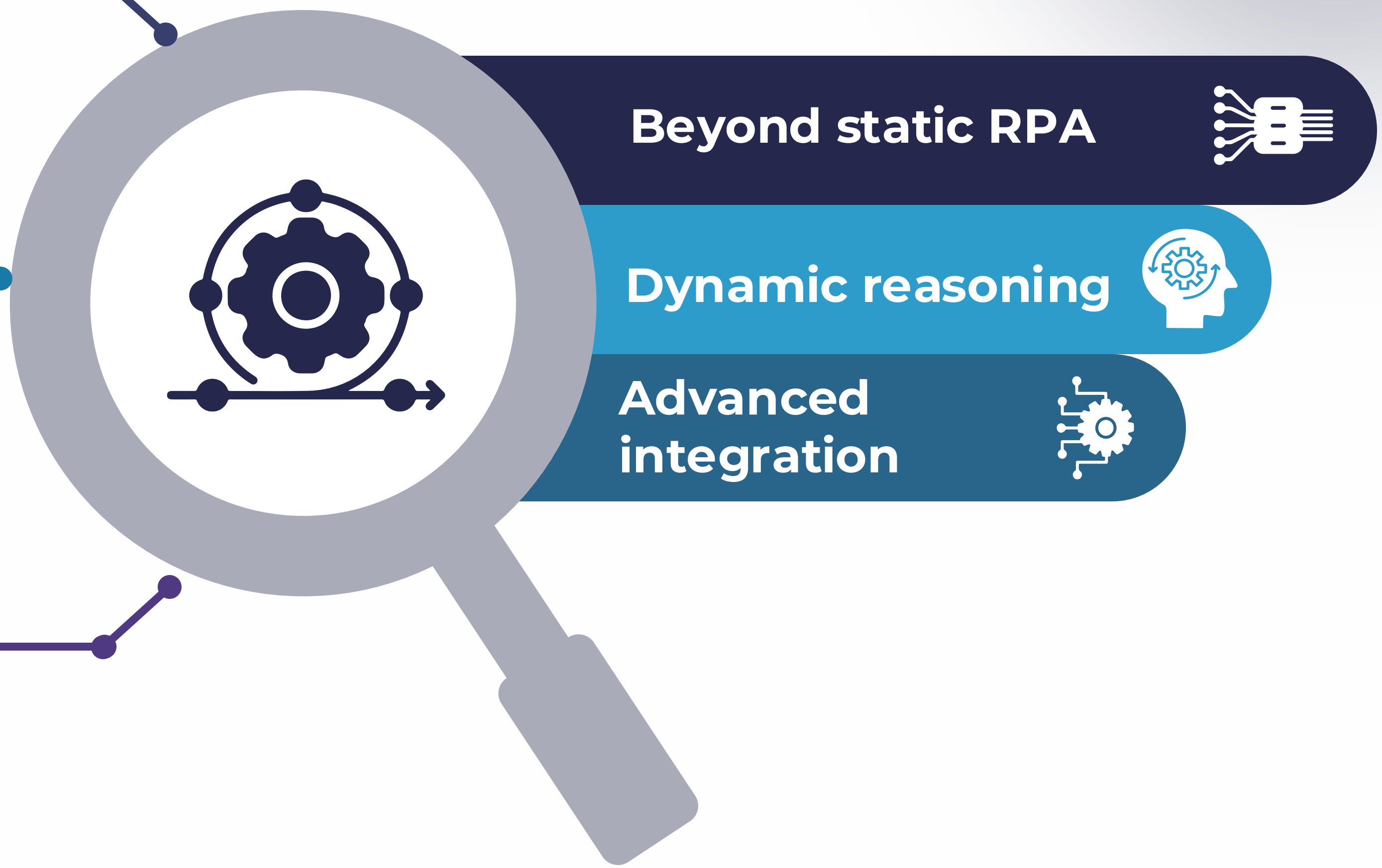
Moving beyond static automation to intelligent orchestration

What Is Workflow Automation?

Evolve from rule-based automation to intelligent orchestration

Enable AI agents to trigger, reason and act based on contextual understanding

Leverage APIs, vector databases and contextual memory systems



Example Pipeline

Email
New support request received

Extraction
Intent parsed and classified

Decision
Resolution action determined

Action
Automated response executed

Feedback
CRM updated, summary sent

Complete end-to-end resolution without manual intervention

Architecture

Input handler
Receive and validate incoming requests

LLM reasoning block
Analyse context and make decisions

Tool executor
Execute API calls and database queries

Memory & feedback store
Learn from interactions

Toolchain Overview

LangChain / Flowise
Agent orchestration and workflow management

Zapier or Make
External trigger integration across platforms

Vector DB
Chroma or Pinecone for semantic search

Monitoring
LangSmith / Evidently for performance tracking

Mini Demo Walkthrough

Scenario: "Order Status Resolution Agent"

Scenario: "Order Status Resolution Agent"

Email reception
Receive customer inquiry in the support inbox

Data extraction
Identify and validate the order ID automatically

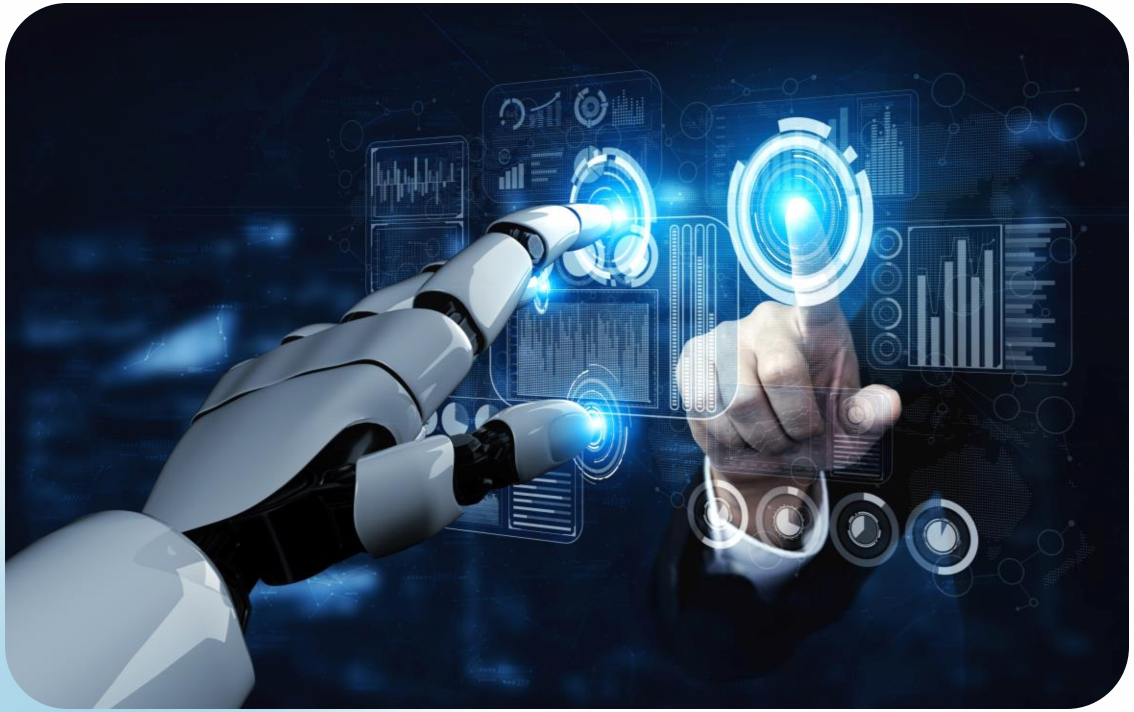
Database query
Retrieve real-time order status from the system

Customer update
Send a personalised response with tracking details

CRM logging
Record the interaction summary automatically in the CRM

Human-in-the-Loop

Strategic Oversight Points



Financial transactions
Require verification before processing payments or transfers

Uncertainty flags
Escalate ambiguous reasoning for human review

Compliance & reliability
Improve accuracy while meeting regulatory requirements

Impact Measurement

Quantifying the Transformation Through Key Performance Indicators

95%

Response time reduction

Customer interactions completed in seconds instead of minutes

3.5×

Task throughput

Workflows completed per hour increased more than threefold

67%

Manual escalation

Far fewer cases require human intervention

100%

Audit trail quality

Every automated action fully documented for compliance and review

Domain-Specific Brainstorm

Time to apply agent thinking to your specific industry and challenges

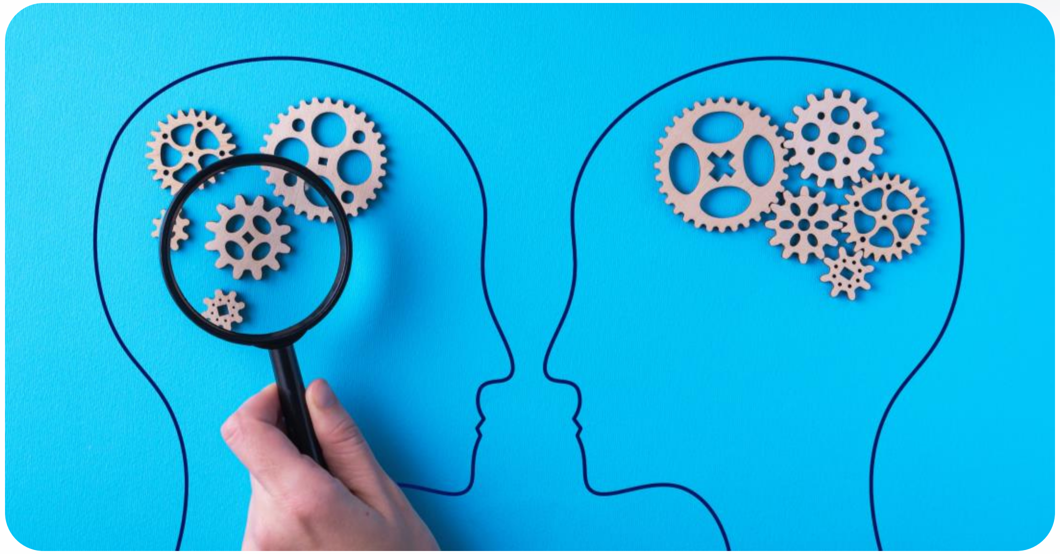
Activity

Identify challenge

Select one high-impact business problem in your domain

Design solution

Ideate how an intelligent agent could automate or enhance the workflow



Ideation Template

Use This Framework to Structure Your Agent Concept

Problem	Data inputs	Agent role	Expected outcome
What task is repetitive or time-intensive?	What information does the agent need?	What actions will it perform?	What measurable improvement do you expect?
Example: Manual order processing	Order forms, inventory data, customer history	Validate orders, check stock, route fulfilment	70% faster processing, fewer errors

Sample Ideas by Domain

Manufacturing

Predictive maintenance coordinator

Monitors equipment sensors, predicts failures, schedules repairs before breakdowns occur

Retail

Dynamic pricing agent

Analyses competitor prices, demand patterns, inventory levels to optimise pricing in real-time

Sample Ideas by Domain

Appointment follow-up assistant

Sends reminders, collects post-visit feedback, flags patients needing follow-up care

Personalised study companion

Adapts learning materials to student pace, answers questions, tracks progress and mastery

Quotation generation assistant

Calculates custom piece pricing based on materials, labour, design complexity automatically

Prioritising Ideas

Evaluation Framework

Use **ROI v. Complexity** matrix to prioritize which agents to build first

	ROI	Complexity
Quick wins	High ROI	Low complexity
Strategic bets	High ROI	High complexity
Avoid	Low ROI	High complexity

Align opportunities with existing system capabilities and data availability

Real-World Case Studies

Proven agent implementations delivering measurable business impact

Case Study: Retail CX Agent

Technology stack
Use the LangChain framework + OpenAI API for natural language understanding

Core capability
Automate customer Q&A by retrieving answers from product manuals and documentation

Business impact
Achieve **35%** lower support costs**, 20%** faster response times and higher customer satisfaction

Multi-Agent System Architecture

Coordinates Routing Decisions and Supplier Scoring in Real-time

Routing agent: Optimises delivery paths based on traffic, weather and priority

Supplier agent: Scores vendors on quality, price and reliability metrics

Inventory agent: Tracks stock levels across warehouses continuously

Key Performance Indicator

18% reduction in shipment delays, improved supplier relationships, better inventory turnover

Case Study: Knowledge Assistant

Training data: Use internal company documents, policies, procedures and historical decisions

User experience: Allow employees to ask questions in natural language and receive instant, accurate answers

Productivity gain: Reduce time spent searching for information by 25%, enabling greater focus on high-value work

Why These Work

High-quality domain data

Clean, structured, relevant training data ensures accurate agent responses and decisions

Strong system integration

Seamless connection to CRM, ERP, databases enables agents to access and act on real information

Continuous monitoring

Regular evaluation, feedback loops and refinement keep agents aligned with business goals

Q&A and Wrap-Up

Let's discuss your ideas and address any questions

Key Takeaways

Scalable intelligence
Agents provide a flexible layer that grows with your business needs

ROI-driven design
Focus on reducing repetitive tasks and increasing speed to create measurable value

Learning from failure
Agent mistakes teach us critical lessons about ethics, governance and system design

Universal potential
Every domain has high-impact use cases waiting to be automated intelligently

Thank you